

Damping versus oscillations in the setting of Hadžić, Rein, Schrecker and Straub

Loop gain, linear theory and particle-in-cell runs

Working notes

3 October 2026, revised 5 and 7 October

Abstract

Hadžić, Rein, Schrecker and Straub proved that, in the radial Vlasov–Poisson system with a central point mass and all particles at the same angular momentum, small steady states $\varepsilon(E_t - E)^k$ damp their linear perturbations if $k > 1$ and do not if $1/2 < k \leq 1$, because a discrete eigenvalue appears below the band of orbital frequencies. The theorem holds below a mass $\varepsilon_0(k)$ that it does not estimate. We reproduce their setting with our tools and study it at finite mass. The criterion for the discrete mode is that the gain λ_{edge} of the self-consistent loop at the lower edge of the band exceeds one; λ_{edge} is the norm of the Mathur operator of their proof. At small mass ($a_0 = 0.01$) $\lambda_{\text{edge}} \approx 0.02$ for $k > 1$ and diverges for $k \leq 1$, as the theorem requires, but the discrete mode of $k \leq 1$ lies 2×10^{-8} band widths below the edge for $k = 0.75$ and about 10^{-98} for $k = 1$. The threshold $\lambda_{\text{edge}} = 1$ is reached at $a_0 = 0.514$ for $k = 1.25$ and $a_0 = 0.920$ for $k = 1.5$; for $k = 2$ and 3 it lies at $a_0 = 1.53$ and 2.38 , so there is no discrete mode even with a shell as massive as the central mass. In these units the coupling parameter η of our earlier work is 1.18 and 2.58 at the first two thresholds, and 5.3 and 10.4 at the other two: η does not mark the transition, λ_{edge} does. The linearized solver confirms all of this in time. Particle-in-cell runs with 10^4 particles follow linear theory to 1% up to $t \approx 1000$ at small mass; at $a_0 = 1$ they show the undamped mode of $k = 0.75$ and $k = 1$ at the predicted frequency (to about 10^{-3}) and the Landau damping of $k = 2$ at the predicted rate (to 5%). The modes that lie just below the edge ($k = 1.25$ and 1.5) lose amplitude in the runs. A scan in amplitude, grid and number of particles shows that this loss is a finite-amplitude effect: it grows with ε and does not change with Δr or N . Three amplitudes with the same 10^5 particles give, for $k = 1.25$, an amplitude within 2.5% of the linear one up to $t = 2000$ with $\varepsilon = 0.01$, a loss of 9% with 0.03 and of 44% with 0.1 . The cause is the resonance of the mode, which lies outside the distribution, at $\Omega_{\min} - \omega_d$ from the edge in frequency: the orbits of the edge are trapped in it when their bounce frequency $\omega_b \propto \sqrt{\varepsilon}$ exceeds half that distance. A pendulum estimate built from the potential of the linear mode, with nothing fitted, gives the threshold ($\varepsilon_c = 0.015$ for $k = 1.25$) and the separatrix that the particles follow; the first minimum of the amplitude is at $\omega_b t \approx 4$. The departure of the reference runs from stationarity has two parts: an initial level set by the grid, and a later growth caused by the discreteness of the particles, which four times more particles remove.

Revision of 7 October 2026. In the first version the values of λ_{edge} for $1 < k < 1.5$ were too low, because the quadrature of the loop gain cut the support of the steady state 3×10^{-8} band widths before its edge (Section 2). With the corrected quadrature, λ_{edge} of $k = 1.25$ at $a_0 = 1$ is 1.583 instead of 1.558 , and the threshold masses are 0.514 and 0.920 instead of 0.53 and 0.93 ; the thresholds of $k = 2$ and 3 and the distance of the modes of $k \leq 1$ to the edge at small mass are now computed. The abstract, Table 2, Figure 1, the values of η at the thresholds and the Discussion were updated. The frequencies of the modes, the linear solutions, the PIC runs and the pendulum estimate do not change. A note on the late-time decay with self-gravity was added to Section 3.3.

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1 Model and theory

1.1 The setting of the theorem

We consider the radial Vlasov–Poisson system in which all particles have the same modulus of angular momentum L_0 , around a point mass $M = 1$ (with $G = 1$),

$$\partial_t F + p_r \partial_r F - \partial_r \Phi_{\text{ef}} \partial_{p_r} F = 0, \quad \Phi_{\text{ef}} = -\frac{1}{r} + \Phi_{\text{self}} + \frac{L_0^2}{2r^2},$$

where Φ_{self} is the potential of the density $4\pi r^2 \rho = 8\pi^2 L_0 \int F dp_r$. This is the system of Hadžić, Rein, Schrecker and Straub [HRSS], with their squared angular momentum $L = L_0^2$. We use $L_0 = 2$, so that the circular orbit of the bare point mass is at $r = 4$ with energy $E_{\text{min}} = -1/8$, and the radial frequency is $\Omega(J) = (J + 2)^{-3}$ without self-gravity.

The steady states are polytropes in the energy,

$$F_{\text{eq}} = A (E_t - E)^k \quad (E < E_t), \quad F_{\text{eq}} = 0 \quad (E \geq E_t),$$

where k is the exponent of the vacuum edge and $E_t = E(J_t)$ is the energy of the orbit of action J_t , the edge of the support. We use $J_t = 0.7$. Without self-gravity this gives $E_t = -0.0686$, within their single-gap condition $-2^{-2/3}/8 = -0.079 < E_t < 0$, which requires $\Omega_{\text{max}} \geq 2\Omega_{\text{min}}$; here $\Omega_{\text{max}}/\Omega_{\text{min}} = 2.46$. The bands of the harmonics $n = 1, 2, \dots$ then overlap, and the only gap in the spectrum is the principal one, $(0, \Omega_{\text{min}})$. The mass a_0 of the shell is measured in units of the central mass; their small parameter ε is proportional to it.

1.2 The theorem

For $k > 1/2$ and ε below a threshold $\varepsilon_0(k)$ [HRSS, Thm. 1.2]:

- if $1/2 < k \leq 1$ the linearized operator has an eigenvalue in the principal gap, so the perturbation does not damp; this includes the King model;
- if $k > 1$ the point spectrum is empty and the field damps in the time-averaged sense $\lim_{T \rightarrow \infty} T^{-1} \int_0^T \|\nabla U\|^2 dt = 0$, without a rate.

The value of $\varepsilon_0(k)$ is not estimated. The proof uses a Birman–Schwinger principle due to Mathur: there is an eigenvalue in the gap if and only if the norm of the Mathur operator M_λ is at least one for some λ in the gap. For $k > 1$ they show $M_\lambda \leq C\varepsilon$, and for $k \leq 1$ that M_λ diverges at the edge of the gap.

1.3 The loop gain

In angle–action variables a mode $\delta F = a(Q, J)e^{-i\omega t}$ obeys, harmonic by harmonic, $(k\Omega - \omega)a_k = kF'_{\text{eq}}(J)\delta\Phi_k(J)$, where $\delta\Phi_k$ is the harmonic of the potential along the orbit J . (Here k is a Fourier index; the edge exponent is also called k , following [HRSS], and the context distinguishes them.) The density sees only the part of δF even in Q , and with $b_k = a_k + a_{-k}$,

$$b_k = R_k \delta\Phi_k, \quad R_k(J; \omega) = \frac{2k^2 \Omega F'_{\text{eq}}}{k^2 \Omega^2 - \omega^2},$$

while the Poisson equation gives

$$\delta\Phi_k = \sum_{k'} \int M_{kk'}(J, J') b_{k'} dJ',$$

$$M_{kk'} = 4\pi L_0 \iint \cos kQ \cos k'Q' G(r(Q, J), r(Q', J')) dQ dQ',$$

with $G = -1/\max(r, r')$. A mode exists if $MR(\omega)$ has the eigenvalue one. For $F'_{\text{eq}} < 0$ and $\omega < \Omega_{\min}$ this operator is similar to the symmetric positive operator $|R|^{1/2}(-M)|R|^{1/2}$; its largest eigenvalue $\lambda(\omega)$ is the gain of the loop potential $\rightarrow$ density $\rightarrow$ potential, and $\lambda_{\text{edge}} = \lambda(\Omega_{\min})$. Since λ grows with ω and $\lambda(0) < 1$ (Antonov),

$$\text{there is a discrete mode below the band} \iff \lambda_{\text{edge}} > 1,$$

and its frequency ω_d is the root of $\lambda(\omega_d) = 1$. $\lambda(\omega)$ is the norm of the Mathur operator with $\lambda_{\text{Mathur}} = \omega^2$. Near the edge $|R_1| \propto (J_t - J)^{k-1}/(\Omega - \omega)$, so λ_{edge} is finite only for $k > 1$. We write $\delta_d = (\Omega_{\min} - \omega_d)/\Delta\Omega$ for the distance of the mode below the edge in band widths. The derivation is in Section 1.5 of the demo document ([docs/demo_eta/demo_eta.pdf](#)).

1.4 The coupling parameter η

In earlier work we used $\eta = \delta\Omega/(\Delta\Omega/\bar{\Omega})$, the self-gravitational shift of the mass-weighted mean frequency, $\delta\Omega = (\bar{\Omega} - \bar{\Omega}_0)/\bar{\Omega}_0$, divided by the relative width of the band; $\bar{\Omega}_0$ is the same average for the same distribution without self-gravity. η needs only the equilibrium. Here we compare it with λ_{edge} .

2 Numerical methods

2.1 Equilibria

The equilibria are built by fixed-point iteration (`equilibrio.py` with `--fondo puntual --forma polE`): the angle-action map of the total potential, the table $E(J)$, the polytrope with $E_t = E(J_t)$ recomputed at each iteration, the density and the Poisson equation, until the potential changes by less than 10^{-13} . The iteration converges in 6–42 iterations for $0.01 \leq a_0 \leq 1$. With the bare point mass the map gives $J(E) = 1/\sqrt{-2E} - 2$ to 1.7×10^{-15} and the circular orbit at $r = 4.000015$. The option is neutral for the isochrone used before: the equilibria, initial states and linear solutions of earlier runs are reproduced bit for bit.

2.2 Loop gain and linear solver

$\lambda(\omega)$ is computed by `lambda_borde.py`: $J = J_t(1 - s^p)$ with 200 Gauss–Legendre nodes in s , 128 points in Q , harmonics up to $k = 6$, and the integrals in Q on a radial grid of 2000 nodes, so that $M = 4\pi L_0 PGP^T$. The exponent is $p = 2$ for $k \geq 1.5$ and $p = 1/(k - 1)$ for $1 < k < 1.5$, which leaves the integrand bounded at the edge, and $p = 4$ for $k \leq 1$. The differences $E_t - E$ and $\Omega - \Omega_{\min}$ are evaluated from the distance to the edge, $J_t - J = J_t s^p$, with the same spline of $E(J)$, so the support of F_{eq} ends exactly at the edge of the band and there is no cancellation in the nodes next to it. In this setting, doubling each resolution changes λ_{edge} by at most 2×10^{-4} for $k = 2$ and 8×10^{-5} for $k = 1.25$ and 1.5 , and ω_d by at most 2×10^{-6} . For $k \leq 1$, λ diverges at the edge; it is computed down to 10^{-12} band widths from it, where doubling the number of nodes changes it by 10^{-5} in relative terms.

Until 7 October the script took E_t from a linear interpolation of the table $E(J)$ and the energies of the nodes from a cubic spline. E_t was then about 10^{-9} too low, and F_{eq} vanished 3×10^{-8} band widths before the edge of the band. That cut rounds off the singularity of the integrand: it lowered λ_{edge} by 0.025 for $k = 1.25$ at $a_0 = 1$ and by 3×10^{-4} for $k = 1.5$, and changed it by less than 10^{-6} for $k \geq 2$; ω_d changed by less than 10^{-8} . The option `borde='tabla'` of the script reproduces the old values bit for bit.

The linearized solver (`lineal.py`) integrates the linear equation in time on 1600×32 nodes in (J, Q) with $\Delta t = 0.5$, from the initial perturbation $\delta F = F_{\text{eq}}(J/J_t)^{3/2} \cos Q$. The observable is

$$h_1(t) = \frac{\sum \delta F B(J) e^{-iQ}}{\sum F_{\text{eq}} B(J)}, \quad B(J) = J^2 e^{-(J-0.35)^2/0.2^2},$$

and $\|\delta\Phi\|$ is the rms value of $\delta\Phi$ over the radii of the shell.

2.3 PIC runs

The PIC code (`VP_PIC`) uses `BGtype = sphere`, a uniform ball of mass 1 and radius 1: for $r > 1$ it is the point mass, and the shells lie at $1.8 < r < 12.2$. Each run has $400 \times 25 = 10^4$ particles on a regular grid in the angle-action variables of the equilibrium, $\Delta r = 0.1$ on $[0, 20]$, the fourth-order Yoshida integrator and $\Delta t = 0.1$ (Courant factor 2). Each perturbed run (D) has a reference run (Z) with the same particles and $\varepsilon = 0$; the perturbation is $(h_{1,D} - h_{1,Z})/\varepsilon$, and the same for $\delta\Phi$. h_1 is computed with the angle-action map of the equilibrium. Table 1 lists the runs.

series	k	a_0	ε	$t_{\text{fin}}, \text{ output}$	purpose
1	0.75, 1, 1.5, 2	0.01	0.1 (and Z)	8000, every 20	the regime of the theorem
2	0.75, 1, 1.25, 1.5, 2	1	0.1 (and Z)	4000, every 5	bound modes and the threshold
3	0.75, 1, 1.5, 2	0.01	0.5 (Z of series 1)	8000, every 20	a lower floor
4	1.25, 1.5	1	0.03, 0.3 (Z of series 2); 0.1 with $\Delta r/2$ or $4N$ (and Z)	4000, every 5	amplitude, grid and particles (Section 3.6)
5	1.25	1	0.01, 0.03, 0.1 (and Z), with 1280×80 particles and $\Delta t = 0.05$	4000, every 5	amplitude with ten times more particles (Section 3.7)

Table 1: The 38 PIC runs. Each takes 220 s to 380 s with four threads, 840 s to 860 s with 4×10^4 particles, and 50 min to 60 min with 1.024×10^5 particles and half the time step (series 5). The energy is conserved to $|\Delta E/E| \leq 1.3 \times 10^{-7}$ at $a_0 = 0.01$ and $\leq 1.8 \times 10^{-5}$ at $a_0 = 1$.

3 Results

3.1 The loop gain and the threshold mass

Table 2 and Figure 1 give λ_{edge} for 36 equilibria.

$k \setminus a_0$	0.01	0.03	0.1	0.3	0.6	1.0
0.75	∞ (2.0×10^{-8})	∞ (1.5×10^{-6})	∞ (1.3×10^{-4})	∞ (0.0044)	∞ (0.027)	∞ (0.075)
1	∞ ($\sim 10^{-98}$)	∞ ($\sim 10^{-33}$)	∞ (4×10^{-11})	∞ (0.00016)	∞ (0.0085)	∞ (0.044)
1.25	0.026	0.076	0.242	0.649	1.122 (0.00013)	1.583 (0.019)
1.5	0.018	0.054	0.170	0.448	0.759	1.051 (0.0015)
2	0.016	0.046	0.145	0.371	0.609	0.820
3	0.016	0.047	0.145	0.358	0.567	0.738

Table 2: λ_{edge} and, in parentheses, the distance δ_d of the discrete mode below the edge in band widths, where there is a mode. For $k \leq 1$, $\lambda_{\text{edge}} = \infty$. The two values marked with $\sim$ are extrapolations with the law $\lambda = a + b \ln(1/\delta)$ from $\delta = 10^{-9}$ and 10^{-12} , where $\lambda = 0.09$ and 0.12 ($a_0 = 0.01$) and 0.27 and 0.36 ($a_0 = 0.03$). Values of 7 October, with the corrected quadrature.

- **The theorem at small mass.** For $k > 1$, λ_{edge} is proportional to the mass, 0.016–0.025 at $a_0 = 0.01$, as their bound $M_\lambda \leq C\varepsilon$ says. For $k \leq 1$, λ diverges at the edge, as $\ln(1/\delta)$ for $k = 1$ and as a power for $k = 0.75$, so there is always a mode. At small mass it is bound by very little: at $a_0 = 0.01$, $\delta_d = 2.0 \times 10^{-8}$ for $k = 0.75$ and of the order of 10^{-98} for $k = 1$.
- **The threshold $\varepsilon_0(k)$.** Solving $\lambda_{\text{edge}} = 1$ with equilibria computed at intermediate masses, the mode appears at $a_0 = 0.514$ for $k = 1.25$ and $a_0 = 0.920$ for $k = 1.5$. For $k = 2$ and 3 there is no mode up to $a_0 = 1$, where $\lambda_{\text{edge}} = 0.82$ and 0.74 ; their thresholds are $a_0 = 1.53$ and 2.38 .

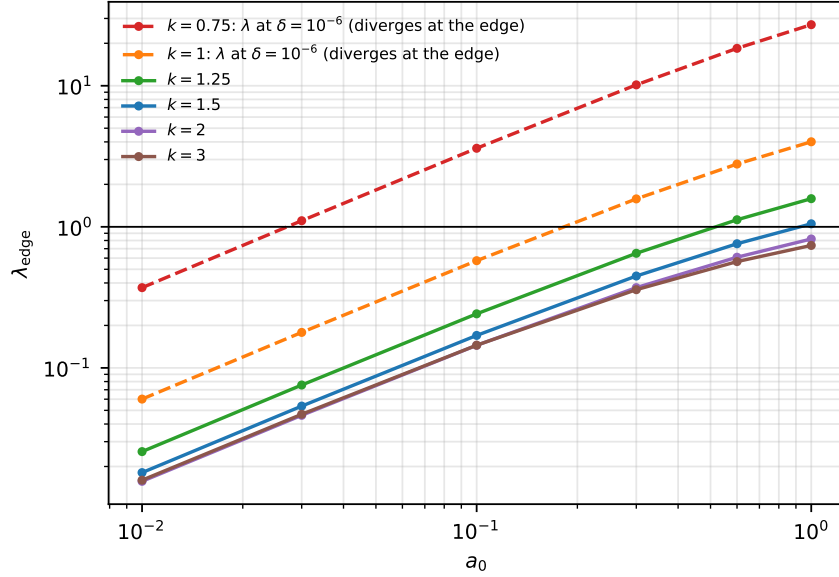


Figure 1: λ_{edge} against the mass for $k > 1$ (solid). For $k \leq 1$, where it diverges, the dashed lines show λ at 10^{-6} band widths below the edge. A discrete mode exists above the horizontal line.

- **Their hypotheses.** $\Omega(J)$ is monotonic in all 36 equilibria. The single-gap condition $\Omega_{\text{max}}/\Omega_{\text{min}} \geq 2$ holds up to $a_0 = 0.3$; it is lost at $a_0 = 0.6$ for $k = 0.75$ (1.99) and at $a_0 = 1$ for $k \leq 1.5$ (1.87–1.98).

3.2 The coupling parameter η

$k \setminus a_0$	0.01	0.03	0.1	0.3	0.6	1.0
0.75	0.015	0.045	0.160	0.558	1.334	2.668
1	0.016	0.048	0.169	0.586	1.388	2.748
1.25	0.017	0.051	0.178	0.613	1.437	2.818
1.5	0.017	0.053	0.187	0.637	1.481	2.883
2	0.019	0.058	0.202	0.680	1.560	2.997
3	0.022	0.066	0.227	0.751	1.686	3.182

Table 3: η of the same equilibria. The relative width of the band is 0.57–0.80, against 0.16–0.29 in the isochrone of our earlier work.

η is 0.015–0.022 at $a_0 = 0.01$, the regime of the theorem (Table 3). At the threshold it is $\eta = 1.18$ for $k = 1.25$ and $\eta = 2.58$ for $k = 1.5$, and $\eta = 3.0$ is not enough for $k = 2$, whose threshold is at $\eta = 5.3$ ($\eta = 10.4$ for $k = 3$). In the isochrone the Wilson model ($k = 2$) had its threshold at $\eta \approx 1.18$. With a band this wide the shape of the edge weighs more in the coupling than the shift of the mean frequency, and η does not mark the transition. The criterion $\lambda_{\text{edge}} = 1$ is the same for every k .

3.3 Linear response in time

At $a_0 = 0.01$ the linear response decays algebraically for every k . The slope of the envelope of $|h_1|$ against $\ln t$ over $1000 \leq t \leq 8000$ is -1.67 , -2.00 , -2.55 and -3.00 for $k = 0.75$, 1 , 1.5 and 2 , close to $t^{-(k+1)}$, the tail of an edge at which the weight vanishes as $(J_t - J)^k$. The sharper the edge, the slower the decay. For $k \leq 1$ the discrete mode is too weakly bound to appear in this time.

(Note of 7 October.) $t^{-(k+1)}$ is the law of the mixing in a fixed potential. With self-gravity the linear solution is that of the mixing with the initial perturbation $\delta F_1(0, J)$ replaced by $\delta F_1(0, J) + iF'_{\text{eq}}(J) \int_0^t \Phi_1(J, t') e^{i\Omega(J)t'} dt'$, and F'_{eq} vanishes at the edge as $(J_t - J)^{k-1}$, one power less. The tail therefore becomes t^{-k} at late times, for $k > 1$. At $a_0 = 0.01$ the crossover is at $t \approx 3.5 \times 10^4$ for $k = 1.5$ and 6×10^4 for $k = 2$, beyond these solutions; for $k \leq 1$ the added term keeps growing and is the precursor of the mode. The derivation and the numbers are in Section 6.C of the notes in `docs/articulo (resultados.py respuesta)`.

At $a_0 = 1$ (Figure 3, black) the solver confirms the loop gain in time. For $k = 0.75, 1$ and 1.25 , $|h_1|$ levels off at $0.075, 0.071$ and 0.059 , and oscillates at $0.14169, 0.14783$ and 0.15330 , against $\omega_d = 0.14169, 0.14785$ and 0.15326 from $\lambda(\omega_d) = 1$. For $k = 1.5$, with the mode 0.0015 band widths below the edge, it decays slowly towards a plateau (0.022 at $t = 4000$) and oscillates at 0.1574 against 0.15751 . For $k = 2$ it is damped: the fitted pole is $\omega = 0.1663\text{--}0.1669$, inside the band, with $\gamma = 6.4\text{--}6.5 \times 10^{-3}$, and $|h_1|$ falls to 10^{-5} by $t = 4000$. After $t \approx 1500$ it follows the tail t^{-k} of the note above: the exponent of its envelope is -2.07 over $2000 \leq t \leq 4000$.

3.4 PIC runs at small mass

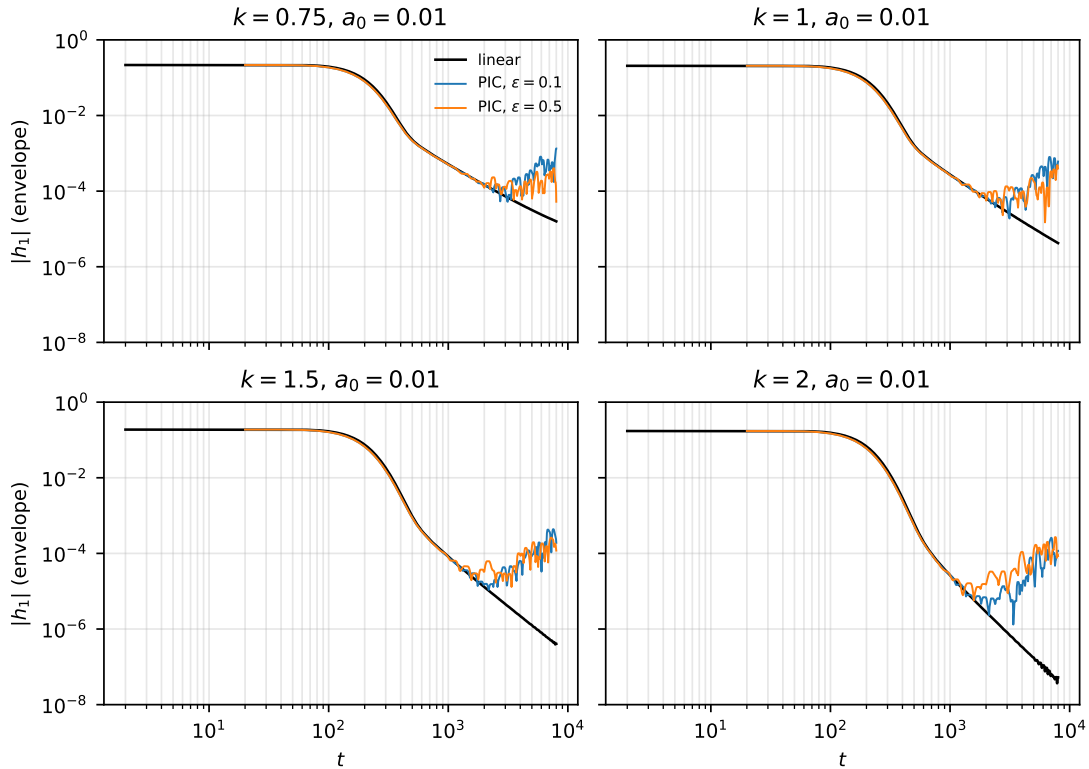


Figure 2: $a_0 = 0.01$: envelope of $|h_1|$ in linear theory (black) and in the PIC runs with $\varepsilon = 0.1$ (blue) and $\varepsilon = 0.5$ (orange).

The PIC runs follow linear theory to 1 % in $|h_1|$ up to $t \approx 1000$ ($12 \tau_1$, with $\tau_1 = 2\pi/\Delta\Omega = 84$), except $k = 2$ (5 %). After that they reach a floor (Figure 2): the D and Z runs decorrelate, and at $t \approx 8000$ their difference is of the order of the noise of the unperturbed run itself. The floor of $(h_{1,D} - h_{1,Z})/\varepsilon$ at $t \approx 4000$ is $(0.2\text{--}1.0) \times 10^{-4}$ with $\varepsilon = 0.1$ and $(1.0\text{--}1.3) \times 10^{-4}$ with $\varepsilon = 0.5$. It does not decrease as $1/\varepsilon$, so it is not subtraction noise: the perturbation itself changes how the noise of the particles evolves. To lower it, more particles are needed, not a larger amplitude. At this mass the runs show the damped edge response of every k ; they cannot show the non-damping of $k \leq 1$.

3.5 PIC runs at $a_0 = 1$

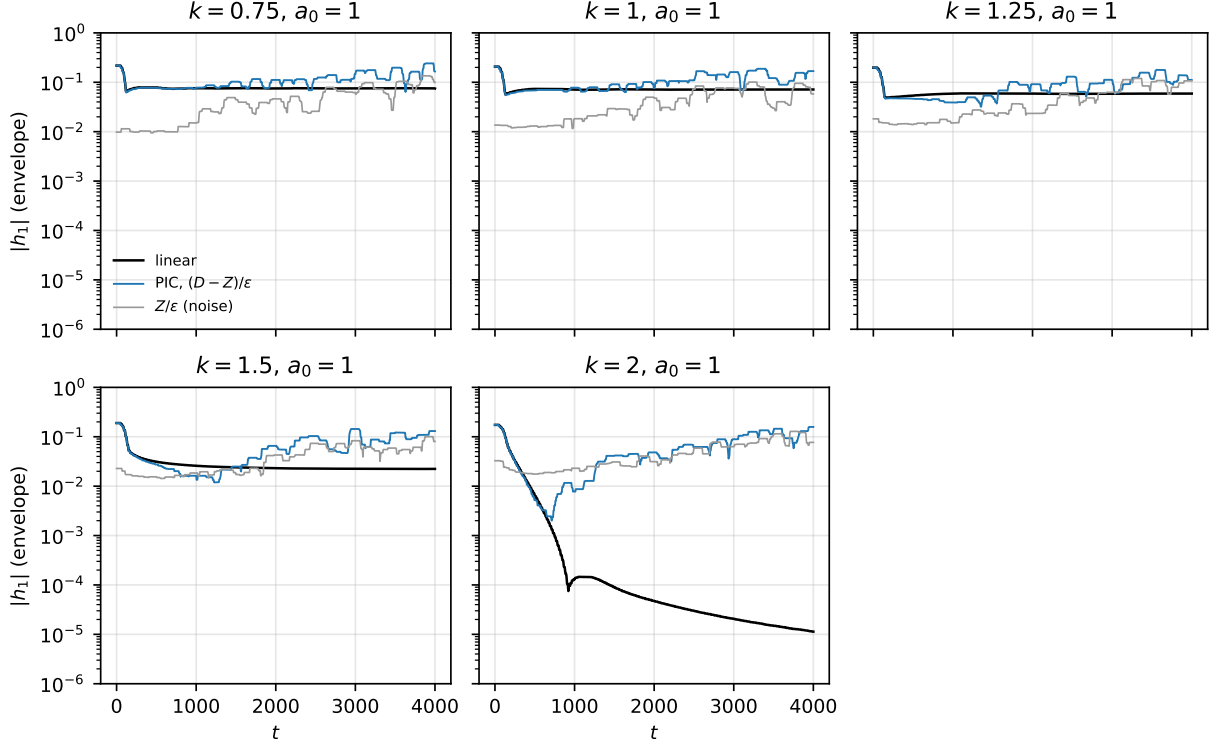


Figure 3: $a_0 = 1$: envelope of $|h_1|$ in linear theory (black), in the PIC runs, $(h_{1,D} - h_{1,Z})/\varepsilon$ (blue), and of the reference run alone, $h_{1,Z}/\varepsilon$ (grey), which measures its departure from stationarity.

k	ω_d ($\lambda = 1$)	window	ω PIC	γ PIC	ω linear	γ linear
0.75	0.14169	300–1500	0.14156	4×10^{-5}	0.14169	4×10^{-6}
1	0.14785	300–1500	0.14766	5×10^{-5}	0.14783	-1×10^{-5}
1.25	0.15326	300–1500	0.15326	4.1×10^{-4}	0.15330	1×10^{-5}
1.5	0.15751	300–1500	0.15805	1.5×10^{-3}	0.15736	1.0×10^{-4}
2	—	100–600	0.16661	6.8×10^{-3}	0.16631	6.5×10^{-3}

Table 4: One-pole fits (matrix pencil, $M = 3$) of h_1 at $a_0 = 1$, before the noise dominates. The fits over 300–1000 (and 100–400 for $k = 2$) give frequencies that differ by at most 5×10^{-4} . For $k = 1.25$ and 1.5 , γ measures a dip of the amplitude, not a decay (Section 3.6).

Table 4 and Figure 3 give the result.

- **$k = 0.75$ and $k = 1$ do not damp.** The PIC amplitude stays within 10% of the linear plateau up to $t \approx 1000$, and the frequency agrees with ω_d to 9×10^{-4} and 1.3×10^{-3} (0.14156 against 0.14169, and 0.14766 against 0.14785).
- **$k = 2$ damps at the linear rate.** The fitted pole agrees with linear theory to 0.2% in ω and 5% in γ , and the PIC amplitude follows the linear one down by a factor of 40, until it reaches the noise at $t \approx 600$.
- **$k = 1.25$ and $k = 1.5$ lose amplitude.** The frequency is that of the discrete mode, but the fitted pole has $\gamma = 4 \times 10^{-4}$ and 1.5×10^{-3} , and the PIC amplitude is 14% to 37% below the linear one at $t = 500$ –1000. These modes lie 0.019 and 0.0015 band widths below the edge. Section 3.6 shows that the loss is a finite-amplitude effect, and that the mode of $k = 1.25$ does not damp.

- **The reference runs are not stationary at this mass.** $|h_{1,Z}|$ grows from 10^{-3} to 10^{-2} between $t = 0$ and 4000, and so does the noise of the subtraction. This limits the comparison to $t \approx 1000\text{--}1500$ ($20\text{--}40 \tau_1$, with $\tau_1 = 37\text{--}48$). Section 3.6 separates the two causes.

3.6 The modes next to the edge

Series 4 tests the loss of the modes of $k = 1.25$ and 1.5 . Each run changes one thing with respect to the run of series 2: the amplitude ($\varepsilon = 0.03$ and 0.3 , with the same reference and the same particles), the grid ($\Delta r = 0.05$ with the same $\Delta t = 0.1$ and the same initial state), or the number of particles ($800 \times 50 = 4 \times 10^4$, which also doubles the resolution of the initial lattice in J and Q). The last two have their own reference. Table 5 gives the PIC amplitude relative to the linear one, and Figure 4 the envelopes.

Series 5 has $1280 \times 80 = 1.024 \times 10^5$ particles for $k = 1.25$, the same ratio of rows in J to points in Q , and $\Delta t = 0.05$. Its three runs, $\varepsilon = 0.01$, 0.03 and 0.1 , share the particles and the reference run, so they differ only in the amplitude. They are the rows “10N” of Tables 5 and 6 and the subject of Section 3.7; they are not in Figure 4.

k	run	$t = 500$	1000	2000	3000	frequency, $t \geq 2500$
1.25	$\varepsilon = 0.03$	0.97 (0.91)	1.21 (0.86)	2.79 (1.33)	4.66 (3.63)	0.2210
	$\varepsilon = 0.1$	0.84 (0.27)	0.66 (0.26)	1.55 (0.40)	1.05 (1.09)	0.2179
	$\varepsilon = 0.3$	0.38 (0.09)	0.47 (0.09)	0.69 (0.13)	0.48 (0.36)	0.1531
	$\Delta r/2$	0.88 (0.08)	0.69 (0.23)	1.44 (0.36)	1.88 (1.26)	0.1998
	$4N$	0.84 (0.25)	0.61 (0.24)	0.81 (0.26)	0.99 (0.27)	0.1530
	10N, $\varepsilon = 0.01$	0.99 (2.55)	0.99 (2.38)	0.99 (2.43)	1.03 (2.43)	0.1532
	10N, $\varepsilon = 0.03$	0.98 (0.85)	0.94 (0.79)	0.91 (0.81)	0.91 (0.81)	0.1531
	10N, $\varepsilon = 0.1$	0.84 (0.25)	0.61 (0.24)	0.79 (0.24)	0.82 (0.24)	0.1531
1.5	$\varepsilon = 0.03$	1.02 (1.62)	1.56 (2.56)	6.69 (6.18)	11.2 (12.4)	0.2094
	$\varepsilon = 0.1$	0.86 (0.49)	0.64 (0.77)	2.39 (1.86)	5.98 (3.71)	0.2117
	$\varepsilon = 0.3$	0.34 (0.16)	0.57 (0.26)	1.83 (0.62)	1.34 (1.24)	0.2283
	$\Delta r/2$	0.86 (0.13)	0.80 (0.49)	2.29 (1.18)	3.89 (3.27)	0.2075
	$4N$	0.84 (0.49)	0.41 (0.56)	0.73 (0.63)	1.34 (0.68)	0.1584

Table 5: Envelope of $|h_1|$ (largest value in $[t, t + 100]$) in the PIC runs divided by the linear one, and in parentheses that of the reference run alone, $|h_{1,Z}|/\varepsilon$, divided by the same linear value. The last column is the frequency of h_1 over $2500 \leq t \leq 4000$. In linear theory it is 0.15325 for $k = 1.25$ and 0.15745 for $k = 1.5$; the band starts at $\Omega_{\min} = 0.15601$ and 0.15774 . The largest value is taken over the stored outputs, every 5 time units in the runs and every 2 in the linear solution. Resampling the series every 0.25 with a quintic spline changes 43 of the 52 ratios by less than 2% and three of them by 7% to 11%, all at $t \geq 2000$.

- **The loss is a finite-amplitude effect.** At $t = 500$ the PIC amplitude is 0.97, 0.84 and 0.38 of the linear one for $\varepsilon = 0.03$, 0.1 and 0.3 ($k = 1.25$), and 1.02, 0.86 and 0.34 ($k = 1.5$). With $\Delta r/2$ or $4N$ it is 0.84–0.88, the value of $\varepsilon = 0.1$. Over $100 \leq t \leq 500$ the difference $|h_1(\varepsilon) - h_1(0.03)|$ is 3.6–7.1 times larger for $\varepsilon = 0.3$ than for $\varepsilon = 0.1$; a correction to h_1/ε linear in ε gives 3.9 and a quadratic one 9.8. Over the same interval, halving Δr or taking $4N$ changes h_1 by 2.8–13 times less than going from $\varepsilon = 0.03$ to 0.1. The demo found the same for its mode next to the edge (D3).
- **The mode of $k = 1.25$ does not damp.** With $4N$ the PIC amplitude is 0.6–1.0 of the linear one up to $t = 4000$, and h_1 oscillates at 0.1530 ($\omega_d = 0.15326$). With $\varepsilon = 0.3$ it falls to 0.38 by $t = 500$ and then stays at 0.5–0.9, also at 0.1531. The $\gamma = 4 \times 10^{-4}$ of Table 4 measures a dip of the amplitude, not a decay. With $\varepsilon = 0.1$ the amplitude reaches a minimum of 0.53–0.63 of the linear one at $t \approx 1350$, with 10^4 or 4×10^4 particles and

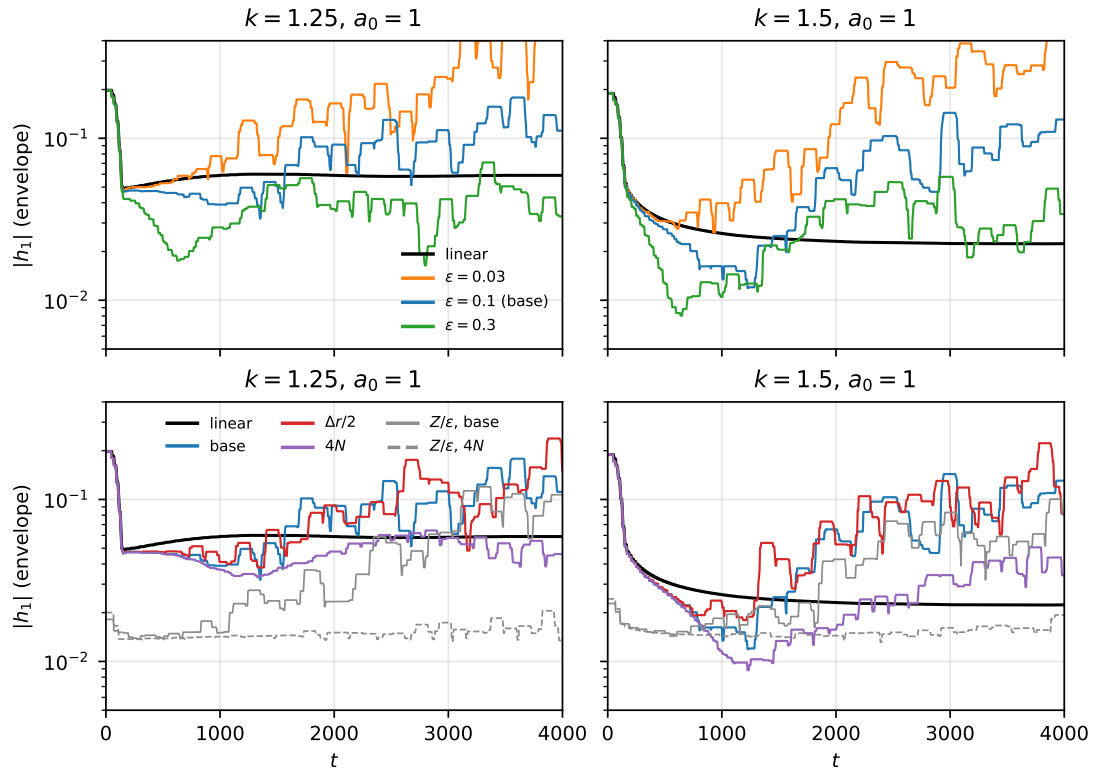


Figure 4: Envelope of $|h_1|$ for $k = 1.25$ (left) and 1.5 (right) at $a_0 = 1$. Top: three amplitudes with the same particles. Bottom: the run of series 2, the same with $\Delta r/2$ and with four times more particles, and the reference runs alone, $|h_{1,Z}|/\varepsilon$ (grey), with 10^4 (solid) and 4×10^4 particles (dashed).

with $\Delta r/2$, and with $4N$ it is back to 1.0 at $t = 2500$. With $\varepsilon = 0.3$ the minimum is deeper and earlier, 0.32 at $t \approx 630$. In the run of series 2 the noise of the reference reaches the signal at $t \approx 2500$ and hides the recovery.

- **With 10^5 particles the run of $\varepsilon = 0.03$ follows the linear solution up to $t = 4000$.** Its envelope stays between 0.90 and 1.14 of the linear one over $100 \leq t \leq 3900$; with 10^4 particles and the same ε it was 1.2 at $t = 1000$ and 4.7 at $t = 3000$. The minimum, 0.90, is at $t \approx 2250$. With $\varepsilon = 0.1$ it was 0.53–0.63 at $t \approx 1350$ and with $\varepsilon = 0.3$, 0.32 at $t \approx 630$: the dip becomes shallower and later as ε decreases. A pole fitted over $300 \leq t \leq 1000$ and over $300 \leq t \leq 1500$ gives the same frequency, 0.15314, with $|\gamma| \leq 3 \times 10^{-5}$; the linear solution gives 0.15330–0.15332 and the loop gain 0.15326. With 10^4 and 4×10^4 particles the fitted frequency changed with the window (0.15292 and 0.15326; 0.15190 and 0.15325). The difference $\max |h_1 - h_1^{\text{lin}}| / \max |h_1^{\text{lin}}|$ is 0.10 over $500 \leq t \leq 1000$ and 0.23 over $1500 \leq t \leq 2000$, of which 0.07 and 0.12 are in the modulus; a frequency lower by 1.6×10^{-4} shifts the phase by 0.16 and 0.32 at $t = 1000$ and 2000. The reference stays at $|h_{1,Z}| = 1.4 \times 10^{-3}$ from $t = 500$ to $t = 3900$, and the rms displacement of the action of its particles is 0.46, 0.41 and $0.56 \times 10^{-3} J_t$ at $t = 1000$, 2000 and 4000, against 1.4, 6.3 and 37 with 10^4 particles and 0.50, 0.58 and 2.5 with 4×10^4 (`relajacion_N.py`).
- **The mode of $k = 1.5$ is not resolved.** It lies $\Omega_{\min} - \omega_d = 2.3 \times 10^{-4}$ below the edge, so it separates from the response of the edge only after $t \sim 1/(\Omega_{\min} - \omega_d) \approx 4300$, longer than the runs; the linear solution itself still decays at $t = 4000$. With $4N$ the PIC amplitude is 0.4–0.7 of the linear one for $1000 \leq t \leq 2000$ and rises to 1.3 by $t = 3000$, at the frequency 0.1584, while the noise of the reference is 0.6–0.7 of the linear amplitude. That is above $\Omega_{\min}$, but the resolution of a window of 1500 is $2\pi/1500 \approx 4 \times 10^{-3}$. These runs do not decide whether the mode survives.
- **The departure of the references from stationarity has two causes** (Table 6). Up to $t \approx 500$, $|h_{1,Z}|$ is $1.4\text{--}2.3 \times 10^{-3}$; with $\Delta r/2$ it is 2.5–4 times less, and with $4N$ it does not change. This is the error of order Δr^2 of the PIC field, which makes the initial state a slightly wrong equilibrium, as in the demo (its Section 5.2). After $t \approx 1000$, $|h_{1,Z}|$ grows to 10^{-2} by $t = 3900$. With $\Delta r/2$ the growth is the same, and with $4N$ it does not appear up to $t = 4000$. This is the discreteness of the particles, as for M5 in the demo.

k	reference	$t = 0$	500	1000	2000	3000	3900
1.25	$10^4, \Delta r = 0.1$	1.8	1.5	1.5	2.3	6.4	11
	$\Delta r/2$	0.72	0.44	1.4	2.1	7.4	12
	$4N$	2.0	1.4	1.4	1.5	1.6	1.7
	$10N$	2.0	1.4	1.4	1.4	1.4	1.4
1.5	$10^4, \Delta r = 0.1$	2.3	1.5	2.0	4.3	8.3	10
	$\Delta r/2$	0.76	0.40	1.2	2.7	7.3	9.4
	$4N$	2.4	1.5	1.4	1.4	1.5	1.9

Table 6: $|h_{1,Z}|$ of the reference runs at $a_0 = 1$, in units of 10^{-3} (largest value in $[t, t + 100]$).

3.7 Amplitude and trapping of the edge orbits

The three runs of series 5 differ only in the amplitude. We compare them with the linear solution through the complex gain

$$\rho(t_0) = \frac{\sum_j W(t_j - t_0) \chi(t_j) \overline{\chi_{\text{lin}}(t_j)}}{\sum_j W(t_j - t_0) |\chi_{\text{lin}}(t_j)|^2}, \quad \chi = \frac{h_{1,D} - h_{1,Z}}{\varepsilon},$$

where χ_{lin} is h_1 of the linear solution, W is a Hann window of total width 200 and t_j are the output times common to the runs and to the linear solution (every 10). $|\rho|$ is the ratio of the amplitudes, and $-d \arg \rho / dt$ is the frequency of the run minus that of the linear solution. The gain is linear in h_1 and does not depend on the sampling: with the outputs every 5 it changes by less than 3×10^{-4} for $t_0 \leq 3000$. The linear solution is not interpolated, because linear interpolation between its outputs has an error of 1.2 %, larger than the difference measured at $t < 250$.

ε	$t_0 = 500$	1000	1500	2000	2500	3000
0.01	0.992	0.990	0.986	0.978	1.004	0.997
0.03	0.976	0.946	0.923	0.908	0.924	0.906
0.1	0.869	0.614	0.591	0.762	0.913	0.830
$\rightarrow 0$, straight line	1.000	1.013	1.020	1.017	1.048	1.048
$\rightarrow 0$, parabola	0.997	1.004	1.017	1.024	1.064	1.063

Table 7: $|\rho(t_0)|$ for the three amplitudes of series 5 ($k = 1.25$, $a_0 = 1$, 1.024×10^5 particles) and for the extrapolations of χ to $\varepsilon = 0$: a straight line through $\varepsilon = 0.01$ and 0.03, and a parabola through the three.

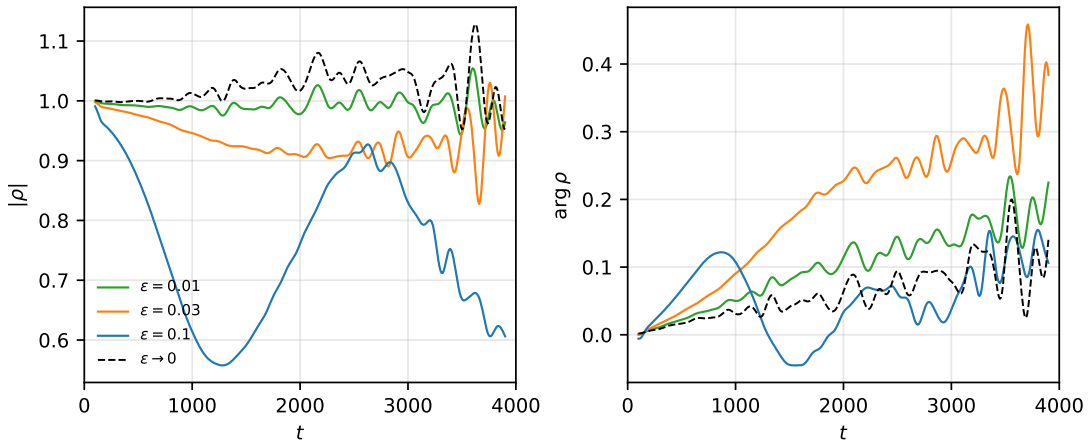


Figure 5: Gain of the PIC response relative to the linear solution for the three amplitudes of series 5: modulus (left) and phase (right). Dashed: the extrapolation to $\varepsilon = 0$ with a straight line.

Table 7 and Figure 5 give the result. With $\varepsilon = 0.01$ the amplitude stays within 0.977–1.005 of the linear one up to $t_0 = 2000$ and within 0.95–1.06 up to $t_0 = 3900$. With $\varepsilon = 0.03$ it falls to 0.907 at $t_0 = 2050$ and stays at 0.89–0.95 up to $t_0 = 3400$. With $\varepsilon = 0.1$ it has a minimum of 0.558 at $t_0 = 1300$, returns to 0.92 at $t_0 = 2650$ and is 0.61 at $t_0 = 3900$. The run of series 2 with the same ε and 10^4 particles gives the same gain for as long as its noise allows: 0.951, 0.871 and 0.629 at $t_0 = 250$, 500 and 1000, against 0.948, 0.869 and 0.614. The ratio $|\rho(0.1) - \rho(0.03)| / |\rho(0.03) - \rho(0.01)|$ is 4.3–5.7 for $t_0 \leq 1000$; a correction to ρ linear in ε gives 3.5 and a quadratic one 11.4. Extrapolated to $\varepsilon = 0$, the amplitude is 1.00–1.05 of the linear one up to $t_0 = 2000$ with the straight line and 0.99–1.06 with the parabola.

Over $250 \leq t_0 \leq 2000$, $\arg \rho$ grows at 5.8×10^{-5} per unit time for $\varepsilon = 0.01$ and at 1.3×10^{-4} for $\varepsilon = 0.03$, and at 2.5×10^{-5} in the extrapolation with the straight line (0.7×10^{-5} with the parabola). The frequency of the PIC response is therefore lower than that of the linear solution by an amount that grows with the amplitude, about $3.5 \times 10^{-3} \varepsilon$, and the two agree to 2.5×10^{-5} at zero amplitude.

The fluctuation of ρ from one window to the next (the rms second difference at steps of 100 in

t_0 , divided by $\sqrt{6}$) is 0.009, 0.003 and 0.005 for $\varepsilon = 0.01, 0.03$ and 0.1 over $500 \leq t_0 \leq 2000$, and 0.04, 0.06 and 0.03 over $2000 \leq t_0 \leq 3900$. With $\varepsilon = 0.01$ the reference run alone is 2.4–2.6 times the linear signal (Table 5); the difference of the two runs recovers the signal with a fluctuation of 1 % up to $t = 2000$.

Bounce frequency. In the angle–action variables of the equilibrium a particle moves with the Hamiltonian $H_0(J) + \varepsilon \delta\Phi(Q, J, t)$, where $\delta\Phi$ is the potential of the perturbation per unit ε , evaluated on the orbit. Let $\Phi_1(J, t) = (2\pi)^{-1} \int \delta\Phi e^{-iQ} dQ$ be its first harmonic in Q . For the discrete mode $\Phi_1 = c_+(J) e^{-i\omega_d t} + c_-(J) e^{i\omega_d t}$, and the part of $\varepsilon \delta\Phi$ that varies slowly on an orbit with $\Omega(J) \approx \omega_d$ is $2\varepsilon|c_+| \cos(Q - \omega_d t + \arg c_+)$. Around the resonant action J_r , where $\Omega(J_r) = \omega_d$, the motion in $\varphi = Q - \omega_d t$ is that of a pendulum,

$$K = \frac{1}{2} \Omega' (J - J_r)^2 + 2\varepsilon|c_+| \cos(\varphi + \arg c_+),$$

with bounce frequency and half-width of the separatrix

$$\omega_b = \sqrt{2\varepsilon|c_+||\Omega'|}, \quad \Delta J_s = \frac{2\omega_b}{|\Omega'|}.$$

Since $\Omega' < 0$, the centre of the island is at $\varphi_O = -\arg c_+$ and the separatrix is $J = J_r \pm \Delta J_s |\cos((\varphi - \varphi_O)/2)|$. The resonance lies outside the distribution, at $J_r = J_t + (\Omega_{\min} - \omega_d)/|\Omega'|$. The island reaches the orbits of the edge when $\Delta J_s > J_r - J_t$, that is, when

$$s \equiv \frac{2\omega_b}{\Omega_{\min} - \omega_d} > 1, \quad \text{or} \quad \varepsilon > \varepsilon_c = \frac{(\Omega_{\min} - \omega_d)^2}{8|c_+||\Omega'|}.$$

An orbit that starts at the edge is then trapped for the phases with $|\cos((\varphi - \varphi_O)/2)| > 1/s$, and it librates up to $J_r + \Delta J_s$ at most. For $s < 1$ it passes below the resonance and reaches $J_t + (J_r - J_t)(1 - \sqrt{1 - s^2})$ at most.

The linear solver stores $\Phi_1(J, t)$ on its rows of J . The step `rebote` of the script fits $c_{\pm}$ on the last row, $J = J_t(1 - 1/3200)$, by least squares over $1000 \leq t \leq 2000$, and takes $\Omega_{\min}$ and ω_d from the loop gain (Table 8). Over $2000 \leq t \leq 4000$ the fit gives a $|c_+|$ that differs by less than 2 % for $k \leq 1.25$ and by 7 % for $k = 1.5$, whose mode is not yet established in the linear solution.

k	$\Omega_{\min} - \omega_d$	$ c_+ $	$ \Omega' $	$\omega_b/\sqrt{\varepsilon}$	ε_c	J_r
0.75	9.90×10^{-3}	7.51×10^{-4}	0.1422	0.01461	0.115	0.7696
1	6.14×10^{-3}	5.76×10^{-4}	0.1475	0.01303	0.056	0.7416
1.25	2.75×10^{-3}	4.06×10^{-4}	0.1520	0.01111	0.0153	0.7181
1.5	2.3×10^{-4}	1.45×10^{-4}	0.1557	0.00672	3×10^{-4}	0.7015

Table 8: Pendulum estimate for the discrete mode at $a_0 = 1$ ($J_t = 0.7$): distance of the mode below the band, first harmonic of the mode potential and $|\Omega'|$ at the edge, bounce frequency, threshold amplitude and resonant action.

The particles. The step `orbitas` follows the particles with $J(0) > 0.6$ in the map of the equilibrium, every 100 time units. Table 9 gives the largest action that they reach, and Figure 6 shows them in the plane $(\varphi - \varphi_O, J)$ near half a bounce period, with the separatrix of the pendulum. Nothing is fitted: ω_d , J_r , ΔJ_s and φ_O come from the loop gain and the linear solver. With $\varepsilon = 0.03$ and 0.1 ($s = 1.40$ and 2.55) the particles of the edge rise along the separatrix and reach $J = 0.747$ and 0.767 ; the estimate gives $J_r + \Delta J_s = 0.743$ and 0.764 . With $\varepsilon = 0.01$ ($s = 0.81$) they stay below the island, at $J \leq 0.710$, against 0.707 from the estimate; in the reference run they reach 0.703 . The mass above $J_t + 0.003$ is at most 1.5×10^{-4} , 1.6×10^{-5} and 1.7×10^{-6} of the total for $\varepsilon = 0.1, 0.03$ and 0.01 .

ε	s	estimate	$t = 500$	1000	1500	2000	3000	4000
0	—	0.7000	0.7020	0.7033	0.7028	0.7016	0.7023	0.7027
0.01	0.81	0.7074	0.7059	0.7093	0.7099	0.7066	0.7056	0.7095
0.03	1.40	0.7434	0.7143	0.7268	0.7401	0.7449	0.7454	0.7443
0.1	2.55	0.7643	0.7454	0.7593	0.7536	0.7573	0.7647	0.7592

Table 9: Largest action of the particles of series 5 at several times, and the value of the pendulum estimate: $J_r + \Delta J_s$ for $s > 1$ and $J_t + (J_r - J_t)(1 - \sqrt{1 - s^2})$ for $s < 1$.

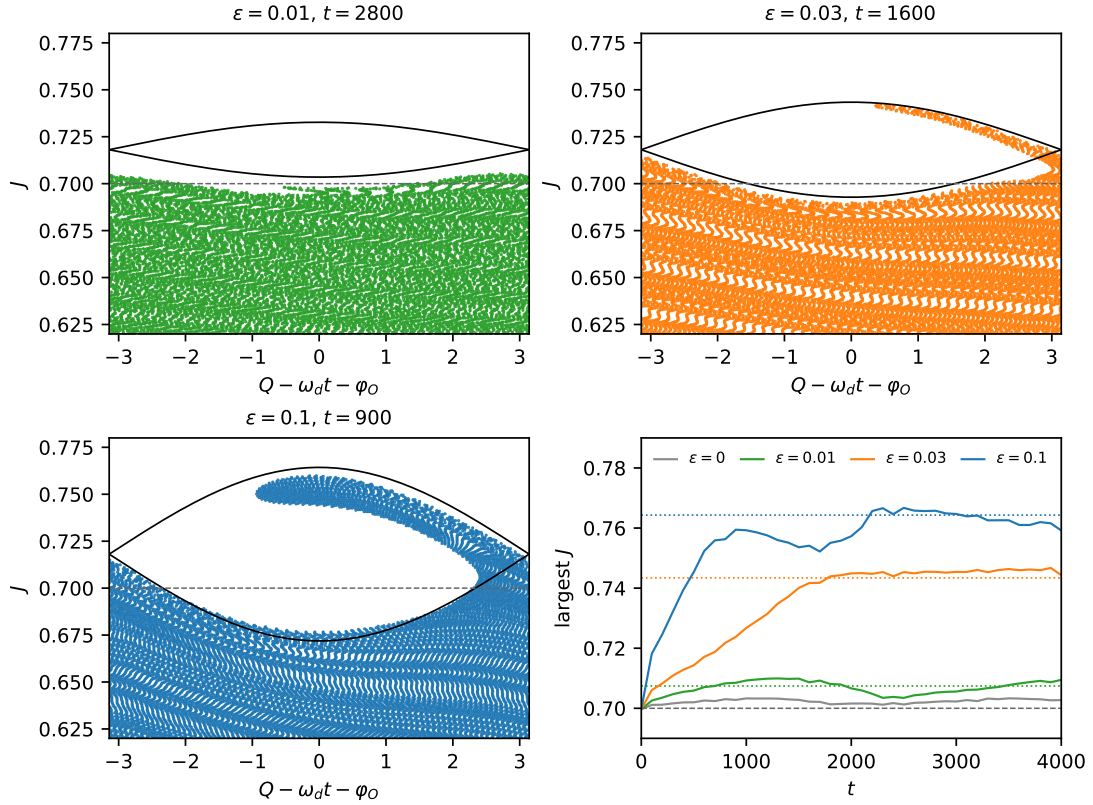


Figure 6: Particles of series 5 with $J(0) > 0.6$ in the plane $(Q - \omega_d t - \varphi_O, J)$ at the output nearest to half a bounce period, $t = \pi/\omega_b$, for the three amplitudes. Black: the separatrix of the pendulum estimate; dashed: the edge J_t . Bottom right: largest action of those particles against time for the three amplitudes and for the reference run (grey); dotted: the values of the estimate.

The runs of series 2 ($\varepsilon = 0.1$, 10^4 particles) test other distances to the band. For $k = 1$ ($s = 1.34$) the particles reach $J = 0.794$ at $t = 1000$ and 0.801 at $t = 1500$, against 0.798 from the estimate. For $k = 0.75$ ($s = 0.93$) they reach 0.728 up to $t = 2000$, against 0.745 : just below $s = 1$ the estimate is steep in s , and $s = 0.80$ would give the measured value. With 10^4 particles the edge particles of the reference runs diffuse to $J = 0.73$ – 0.75 by $t = 3000$, so these comparisons end at $t = 2000$.

Time and depth of the loss. The first minimum of $|\rho|$ is at $\omega_b t_0 = 3.9$ for $\varepsilon = 0.03$ and at 4.6 for $\varepsilon = 0.1$ in series 5, and at 4.0 for $\varepsilon = 0.1$ and 0.3 with 10^4 particles. With $\varepsilon = 0.1$ the following maximum is at $\omega_b t_0 = 9.3$. Table 10 gives $|\rho|$ of the runs with a mode at equal values of $\omega_b t_0$. The four runs of $k = 1.25$ with $\varepsilon = 0.1$ have 10^4 , 4×10^4 and 1.024×10^5 particles, $\Delta r = 0.1$ and 0.05 , and $\Delta t = 0.1$ and 0.05 ; they agree to 0.01 at $\omega_b t_0 = 1$ and to 0.03 at $\omega_b t_0 = 2$ and 3 . The loss is ordered by s . Two runs with different k and ε and nearly the same s ($k = 1$ with $\varepsilon = 0.1$, and $k = 1.25$ with $\varepsilon = 0.03$) give 0.939 and 0.944 at $\omega_b t_0 = 2$, and 0.935 and 0.920 at $\omega_b t_0 = 3$.

k	ε	run	s	$\omega_b t_0 = 1$	2	3	4
1.25	0.01	10N	0.81	0.985	1.005	0.992	1.055
0.75	0.1	series 2	0.93	0.967	0.988	0.985	0.988
1	0.1	series 2	1.34	0.956	0.939	0.935	0.977
1.25	0.03	10N	1.40	0.975	0.944	0.920	0.911
1.25	0.1	series 2	2.55	0.943	0.838	0.677	0.520
		$\Delta r/2$		0.949	0.864	0.705	0.610
		4N		0.941	0.835	0.681	0.568
		10N		0.940	0.838	0.684	0.570
1.25	0.3	series 4	4.42	0.866	0.667	0.419	0.310

Table 10: $|\rho(t_0)|$ at equal times in units of the bounce time. The value $\omega_b t_0 = 4$ corresponds to $t_0 = 866, 971$ and 1139 in the runs of series 2 and to $t_0 = 3602$ in the first row, where the noise of those runs is no longer small. The run of $k = 1.25$ with $\varepsilon = 0.03$ and 10^4 particles is omitted: its noise exceeds the loss.

ε_c is proportional to $(\Omega_{\min} - \omega_d)^2$, so the amplitude that a run may use falls quickly as the mode approaches the band: $0.115, 0.056$ and 0.015 for $k = 0.75, 1$ and 1.25 , and 3×10^{-4} for $k = 1.5$. With $s = 0.81$ the loss was within the noise and with $s = 1.4$ it was 9% . For $k = 1.5$, $s = 0.8$ means $\varepsilon = 2 \times 10^{-4}$, fifty times less than the smallest amplitude run here, for which the reference alone is already 2.5 times the signal.

4 Discussion

Established.

- The criterion $\lambda_{\text{edge}} > 1$, the norm of the Mathur operator of [HRSS] at the edge of the gap, reproduces their dichotomy at small mass: $\lambda_{\text{edge}} \propto a_0$ for $k > 1$ and $\lambda_{\text{edge}} = \infty$ for $k \leq 1$.
- At finite mass it gives the threshold that the theorem does not estimate: $a_0 = 0.514$ for $k = 1.25$, $a_0 = 0.920$ for $k = 1.5$, and above 1 for $k \geq 2$ (1.53 for $k = 2$ and 2.38 for $k = 3$).
- The time-domain linear solver agrees with the loop gain: plateaus at the frequency ω_d , to 4×10^{-5} for $k \leq 1.25$, where $\lambda_{\text{edge}} > 1$, and damping where it is below one.
- The PIC runs reproduce the undamped modes of $k \leq 1$ and the damping of $k = 2$ at $a_0 = 1$.

- The loss of amplitude of the modes next to the edge in the PIC runs is a finite-amplitude effect: it grows with ε and does not change with Δr , Δt or N . With 4×10^4 particles the mode of $k = 1.25$ persists up to $t = 4000$ at ω_d . With the same 1.024×10^5 particles its amplitude stays within 0.977–1.005 of the linear one up to $t = 2000$ for $\varepsilon = 0.01$, falls to 0.91 for $\varepsilon = 0.03$ and to 0.56 for $\varepsilon = 0.1$; extrapolated to $\varepsilon = 0$ it is 1.00–1.05. The frequency is lower than the linear one by about $3.5 \times 10^{-3}\varepsilon$.
- The loss comes from the resonance of the mode, which lies outside the distribution at $\Omega_{\min} - \omega_d$ from the edge. The pendulum estimate of Section 3.7, with nothing fitted, gives the threshold $s = 2\omega_b/(\Omega_{\min} - \omega_d) = 1$, the separatrix that the particles of the edge follow and the largest action that they reach when trapped (to 0.004, for $k = 1$ and 1.25). The first minimum of the amplitude is at $\omega_b t \approx 4$, and the loss at equal $\omega_b t$ is ordered by s across k and ε .
- The reference runs depart from stationarity for two reasons: the error of order Δr^2 of the PIC field sets the initial level, and the discreteness of the particles makes it grow after $t \approx 1000$. Four times more particles remove most of the growth up to $t = 4000$, and ten times more remove it.
- η does not mark the transition in this setting.

Open.

- At small mass the discrete mode of $k \leq 1$ is bound by 2×10^{-8} band widths ($k = 0.75$, $a_0 = 0.01$) or far less ($k = 1$), so a finite simulation sees only a slowly decaying edge response. The non-damping that the theorem proves is not observable there; it becomes observable only at large mass.
- The mode of $k = 1.5$ at $a_0 = 1$, bound by 2.3×10^{-4} in frequency. Deciding whether it survives in the PIC runs needs $t \gtrsim 10^4$, the noise of $4N$ or less and, by Section 3.7, an amplitude below $\varepsilon_c = 3 \times 10^{-4}$.
- The depth of the loss. The pendulum estimate gives the threshold and the scale of time, not the amplitude of the mode once the edge orbits are trapped; Table 10 only tabulates it against s .
- The estimate for s just below one. For $k = 0.75$ with $\varepsilon = 0.1$ the particles of the edge reach $J = 0.728$, against 0.745. The estimate uses c_+ and Ω' at the edge, and there the resonance is $J_r - J_t = 0.07$ beyond it.
- The noise floor at small mass, which limits the comparisons to $\sim 10^3$ time units with 10^4 particles. Lowering it needs more particles; at $a_0 = 1$, 4×10^4 particles extend the comparison to $t = 4000$.

5 Reproduction

Everything is produced by `reproducir/scripts/hadzic.py` from that directory:

step	what it does	time
lineal	the 36 equilibria, λ_{edge} (<code>exe/hadzic/lineal/lambda.txt</code>), η (<code>eta.txt</code>) and the linear solutions of the runs	40 min
resultados.py umbrales	(another script of the same directory) the threshold masses, solving $\lambda_{\text{edge}} = 1$ with equilibria at intermediate masses (<code>exe/resultados/umbrales.txt</code>)	8 min
rebote	bounce frequency of the edge orbits in the mode of each k at $a_0 = 1$ (<code>exe/hadzic/lineal/rebote.txt</code>); needed by <code>analizar</code> , <code>orbitas</code> and <code>figuras</code>	2 min
preparar	initial states and <code>.par</code> files in <code>reproducir/corridas/13_hadzic/</code>	10 min
correr	the 38 PIC runs, one after another, with four threads	7 h
serie	h_1 and $\delta\Phi$ of the named runs, with the snapshots split among several processes; used for the four runs of series 5	2 h
analizar	h_1 and $\delta\Phi$ in the map of the equilibrium, envelopes, slopes, pole fits, the table of series 4 and 5, the gain of the amplitude scans and the loss in units of the bounce time (<code>exe/hadzic/resumen.txt</code>)	3 h
orbitas	action of the particles of the edge in series 5 and in three pairs of series 2 (<code>exe/hadzic/orbitas.txt</code>)	1 min
figuras	the figures of this report	1 min
videos	one video per perturbed run, in <code>exe/hadzic/videos/</code> : the particles in (r, p_r) , the harmonic $k = 1$ of the perturbation in (Q, J) and $ h_1 $	3 h

The runs of series 1 used the Courant factor 2 of the template, so they reached $t = 8000$; their `.par` files are kept as used, and their header comment says $t_{\text{fin}} = 4000$. The runs with $\Delta r/2$ use the initial states of series 2; the dictionary CAMBIOS of the script lists the changes of each run of series 4 and 5.

References

- [HRSS] M. Hadžić, G. Rein, M. Schrecker, C. Straub, *Damping versus oscillations for a gravitational Vlasov–Poisson system*, Arch. Ration. Mech. Anal. 249:45 (2025), [arXiv:2301.07662](#).
- [HRS] M. Hadžić, G. Rein, C. Straub, *On the existence of linearly oscillating galaxies*, Arch. Ration. Mech. Anal. 243 (2022), [arXiv:2102.11672](#).
- [S] C. Straub, *Numerical experiments on stationary, oscillating, and damped spherical galaxy models*, [arXiv:2405.01235](#) (2024).

The other works of the group and the comparison of their setting with ours are in `docs/hadzic/hadzic_base.md`.